Atlanta, GA
(813) 215-7748
nsciccarello@gmail.com
www.github.com/nsciccarello
www.linkedin.com/in/nicoleswatson

Nicole Ciccarello

Data Engineer II

SUMMARY

I am an Atlanta-based Data Engineer skilled in creating and optimizing data pipelines with Python, SQL, Spark, and Scala. I am also an expert in cloud technologies like AWS and Azure, specializing in performance optimization and ETL automation. With my unique background in journalism, my enhanced analytical and storytelling abilities enable me to deliver impactful data solutions for major clients like Versant Media, NBCUniversal, and The Coca-Cola Company.

EXPERIENCE

Data Engineer II

January 2024 - Present

SportsEngine - PlayMetrics

Remote

Previously part of NBCUniversal and Versant Media.

- Designed and implemented scalable ETL pipelines and data workflows, reducing runtime by 30%.
- Coordinated with cross-functional teams to ensure data accuracy and reliability, supporting key business decisions.
- Streamlined data operations using Python, Amazon MWAA (Airflow), AWS, and SQL, reducing manual tasks by 40%.

HIGHLIGHTED PROJECT:

Soccer Shots Project

- Engineered 25+ Airflow DAGs spanning 43 datasets for data ingestion and transformation, reducing processing time by 57%.
- Delivered GPG-encrypted exports through AWS S3 across 4 schedules, supporting financial reporting and partner integrations.
- Optimized 6 large datasets using 5,000-record pagination and read-replica routing to limit database contention.
- Implemented checkpoint-based recovery across 43 datasets, automatically backfilling missed export windows.
- Corrected timezone handling to eliminate 7–8 hours of redundant overlap per hourly export run.
- Automated daily Salesforce updates for 14 financial metrics, supporting account retention and revenue expansion.

Data Engineer

August 2022 - January 2024

The Coca-Cola Company

Atlanta, GA

- Deployed 2 new ETL pipelines using Scala, Java, Spark, and SQL to help automate data processing from legacy systems.
- Optimized algorithm and metric processing, reducing runtime by 25%.
- Migrated 7 PB of data, along with stored procedures, data dictionaries, and tables, from SQL Server to Synapse Analytics.

HIGHLIGHTED PROJECTS:

Global Supply Chain Project

- Developed a global data pipeline for 250+ Coca-Cola bottlers worldwide using Scala, Spark, SQL, and Microsoft Azure.
- Reduced monthly workload by 63% by automating the ETL process and efficiently handling billions of rows of data.
- Implemented a Spark-based analytics solution for large-scale datasets, improving data analysis time by 50%.

Marketing Media Project

- Created ETL pipelines handling data from more than 50 different databases, supporting marketing clients such as Google.
- Enhanced performance by 40% by converting SQL Server stored procedures to Spark SQL for Synapse Analytics integration.
- Successfully migrated 3PB of data to Azure Data Lake Storage (ADLS), optimizing storage costs and data accessibility.

Software QA Engineer - Data Warehouse

March 2019 - July 2022

Warner Bros. Discovery

Atlanta, GA

- Collaborated closely with data engineers, acquiring advanced data engineering skills to improve project results.
- Identified and reported 98% of software bugs in testing phases, significantly enhancing product quality and reliability.
- Implemented data storage and warehousing solutions using AWS S3 and Redshift, ensuring data integrity and scalability.

HIGHLIGHTED PROJECT:

CNN Library Migration Project

- Spearheaded tests to migrate petabyte-scale CNN digital content to AWS S3 for 2,000+ employees using Java and Selenium.
- Automated daily regression tests with Java and IntelliJ IDEA, reducing test duration by 60-75%.
- Ensured data integrity and accuracy during cloud migration, employing rigorous testing protocols and tools.
- Utilized AWS CloudWatch for monitoring and logging test environments, enabling proactive issue detection and resolution.

OTHER TECHNICAL CONTRIBUTIONS

Data Engineer Fellow

March 2022 - August 2022

Correlation One - Data Engineering Fellowship

Remote

- Selected for a part-time fellowship (4% acceptance rate), with learning driven by real-world business cases.
- Developed expertise in AWS S3, Redshift, and Lambda to optimize storage, processing, and ETL automation.
- Mentored junior data engineers during the project, promoting knowledge sharing and professional growth.

HIGHLIGHTED PROJECT:

Movie Recommendations for Domestic and International Flights

- Directed a team of 6 to construct an ETL pipeline to help airlines select in-flight entertainment for passengers.
- Architected an airline ETL pipeline using Python, SQL, Hadoop, PySpark, Hive, Amazon ECRs, and APIs, processing 1M+ flight records.
- Received the 'Most Distinguished Project' award, recognizing exceptional innovation and technical proficiency.

Founder and Data Engineer

July 2021 - July 2024

Nicky Loves Data

Atlanta, GA

- Founded a data engineering and analytics blog to showcase projects and skills such as data transformation and data visualization.
- Analyzed Super Bowl game data from CSV files to study correlations between outcomes, viewership, and halftime shows.
- Discovered with 95% confidence that Super Bowl viewership will decrease due to large team score differences.

TECHNICAL SKILLS

Programming Languages: Python, Scala, SQL, Java, R, JavaScript

Cloud Technologies: AWS, Microsoft Azure

- **AWS Tools:** S3, Glue, Redshift, MWAA (Apache Airflow), EMR, RDS, Lambda, Athena, DynamoDB, IAM, ECR, DMS
- **Microsoft Azure Tools:** Synapse Analytics, Data Lake Storage, Blob Storage, SQL Database, Data Factory, Databricks

Big Data Technologies: Microsoft SQL Server, Apache Spark, Hadoop/HDFS, Hive, Apache Airflow

Data Engineering Skills: Data Warehousing, Data Modeling, Data Normalization, Data Processing, ETL Development, Distributed Computing

DevOps & Version Control: Git, GitHub, GitLab, Jenkins, Jira

Other Technologies: Sisense, REST APIs, Selenium, Power BI, Excel

EDUCATION & CERTIFICATIONS

Western Governors University - Remote

Master of Science in Data Analytics – Data Engineering

Correlation One Data Skills for All - Remote

Data Engineering Fellowship Certificate - Honors

Harvard University - Remote

Courses in Computer Software Engineering / Computer Science

University of South Florida - Tampa, FL

B.A. in Mass Communications